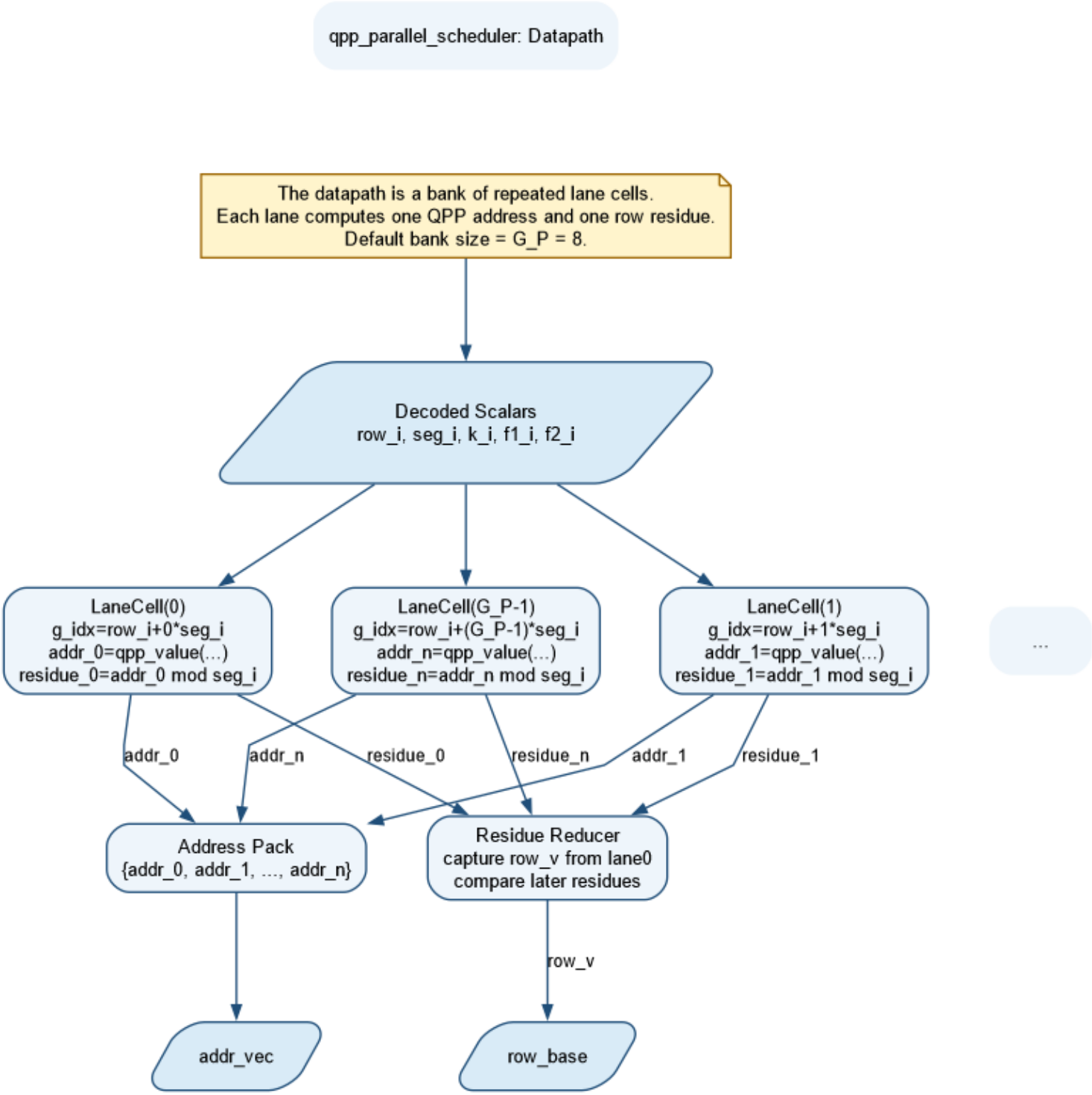
Isomorphic Architecture Album

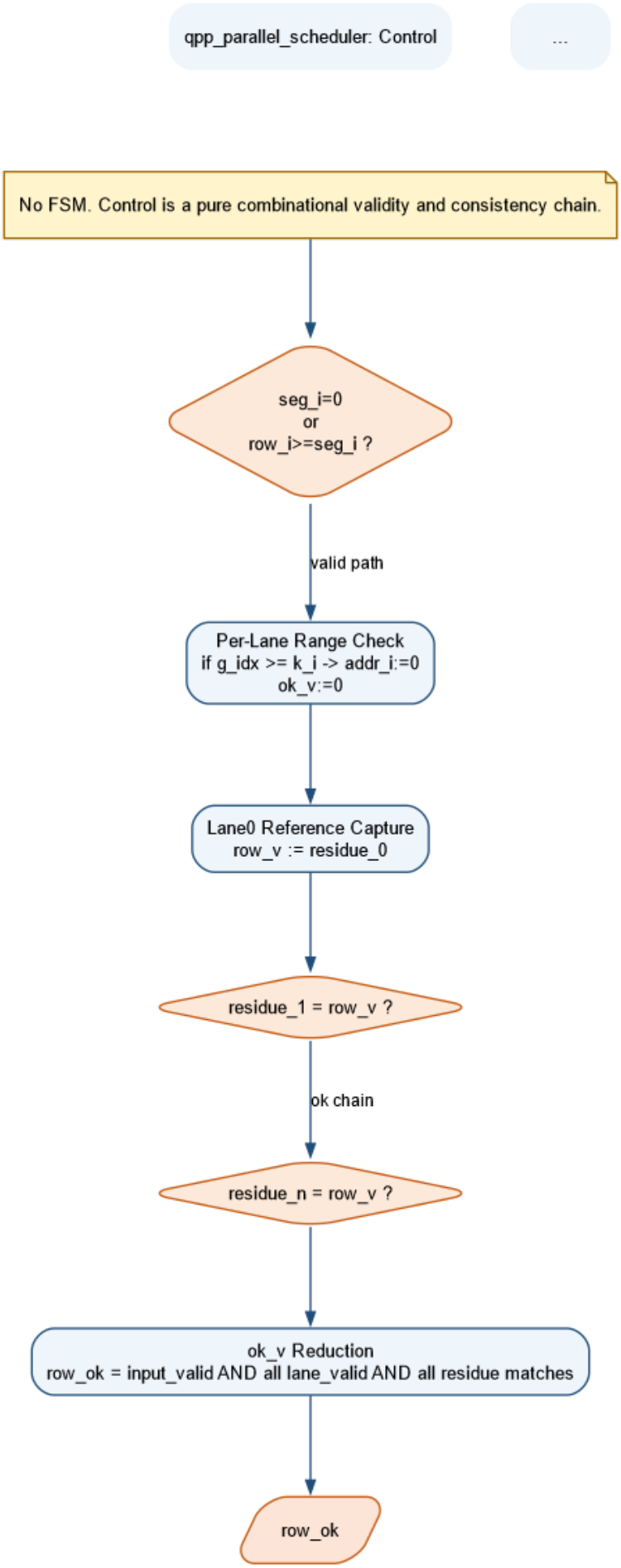
QPP Parallel Scheduler

Datapath



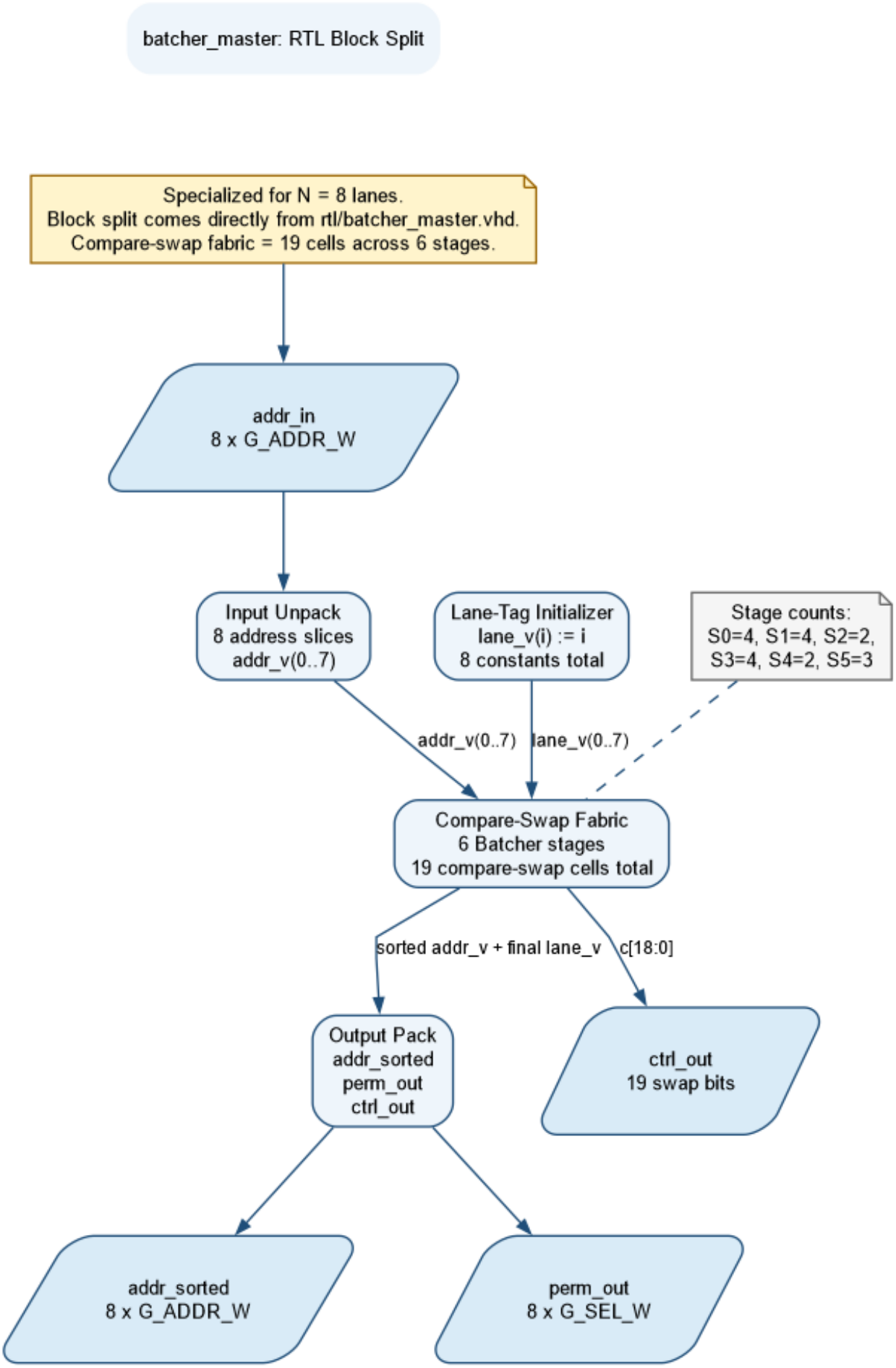
QPP Parallel Scheduler

Control Logic



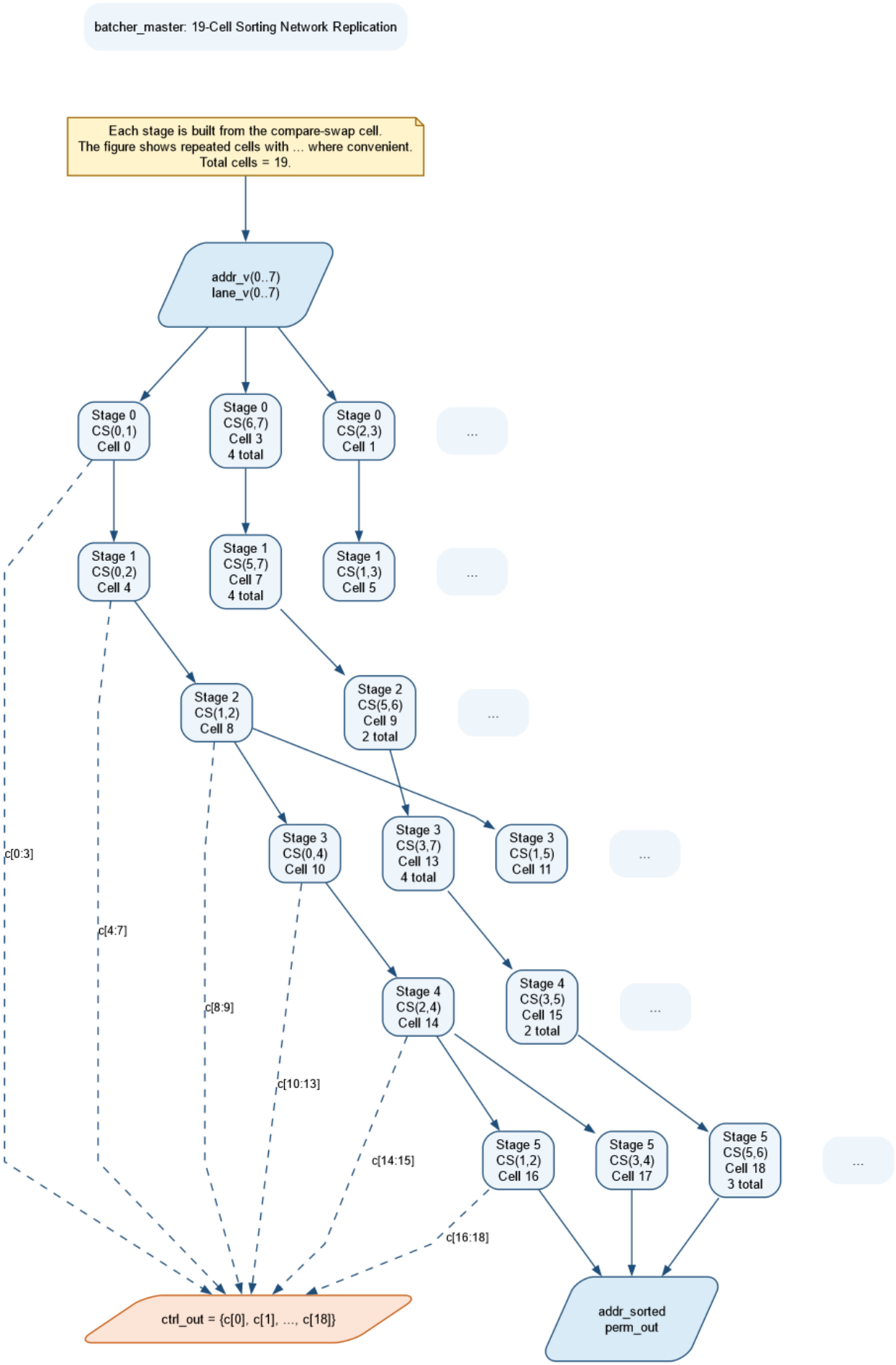
Batcher Master

Structural Split



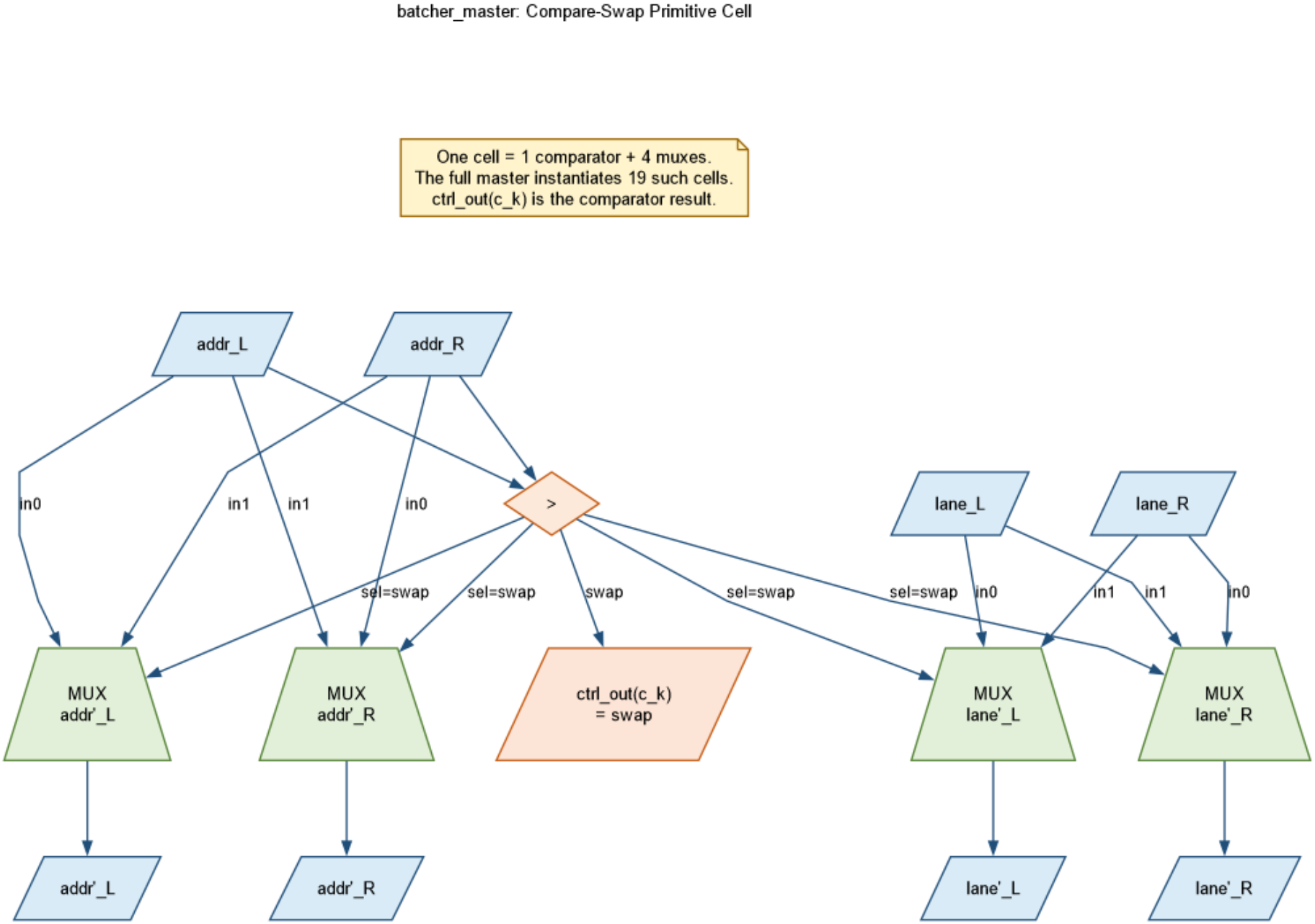
Batcher Master

Sorting Network Datapath



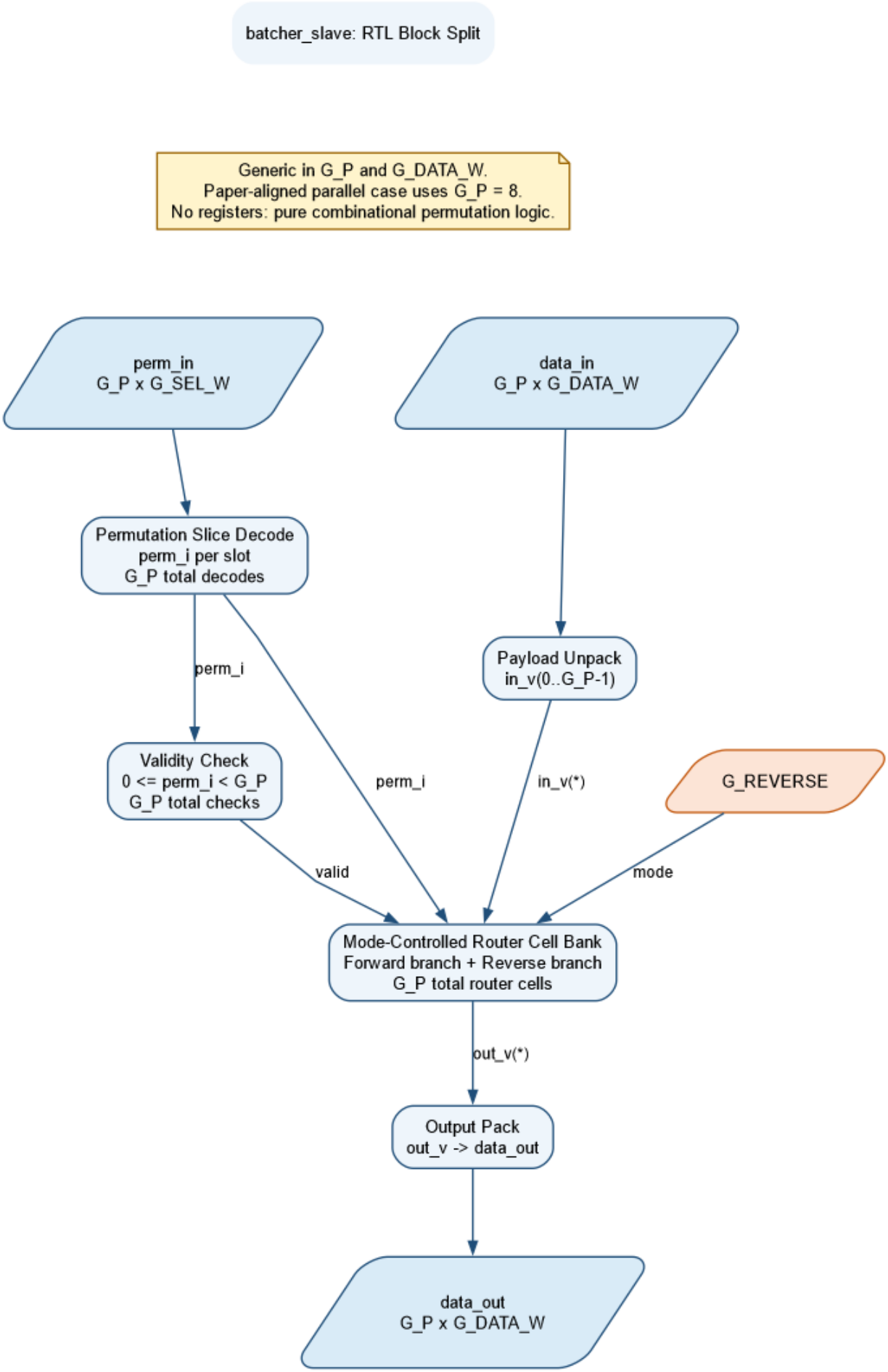
Batcher Master

Compare-Swap Cell



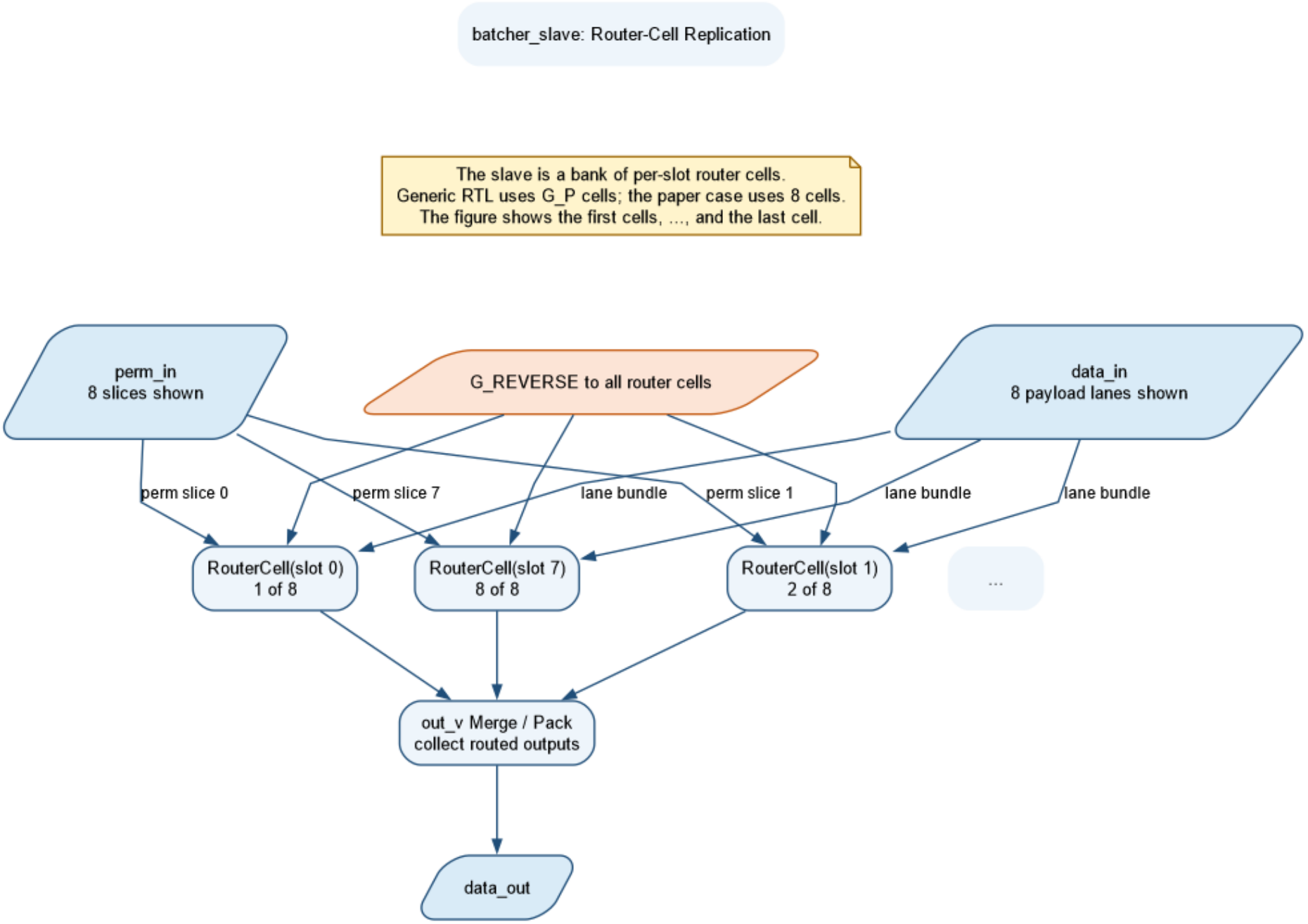
Batcher Slave

Structural Split



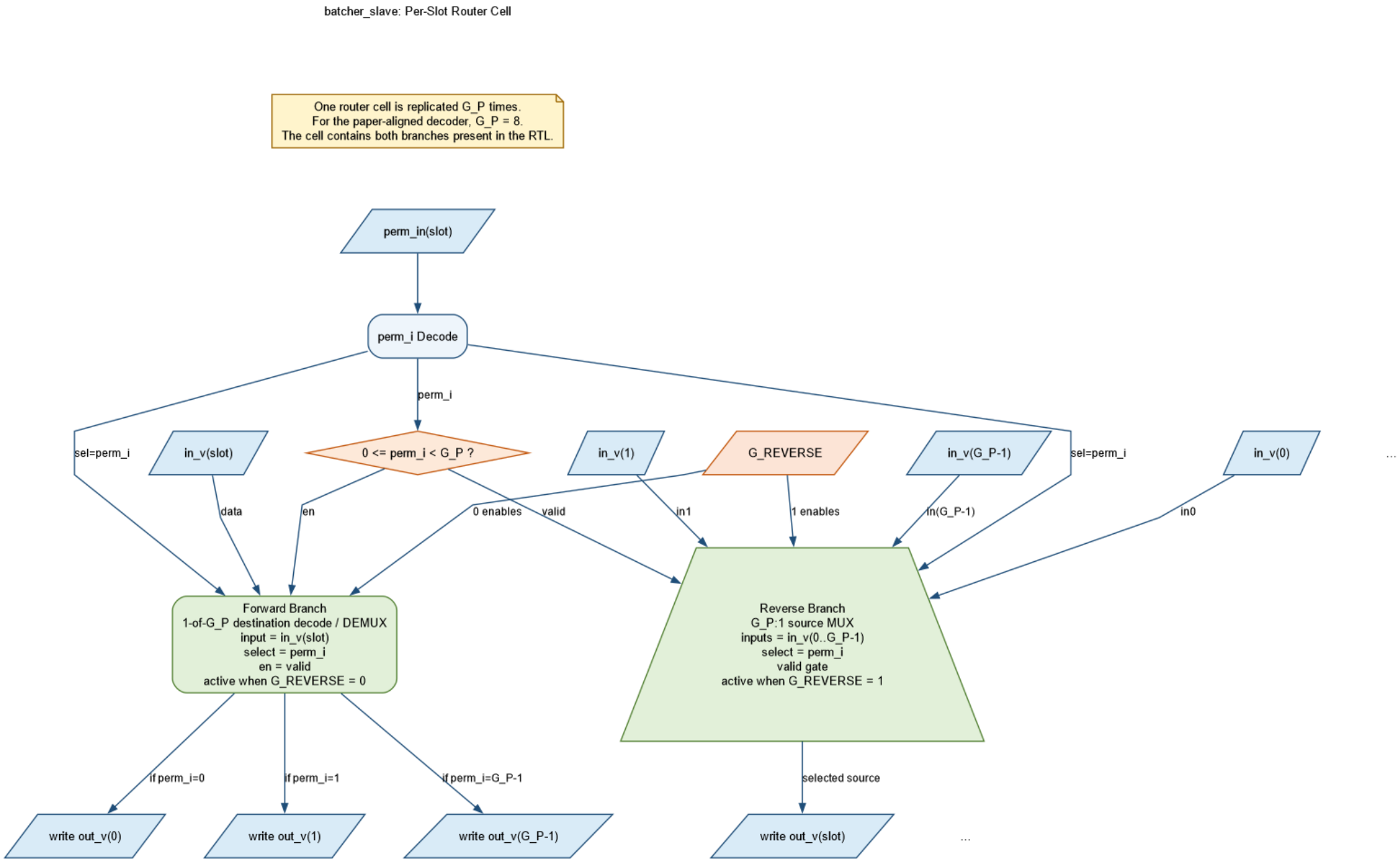
Batcher Slave

Permutation Network



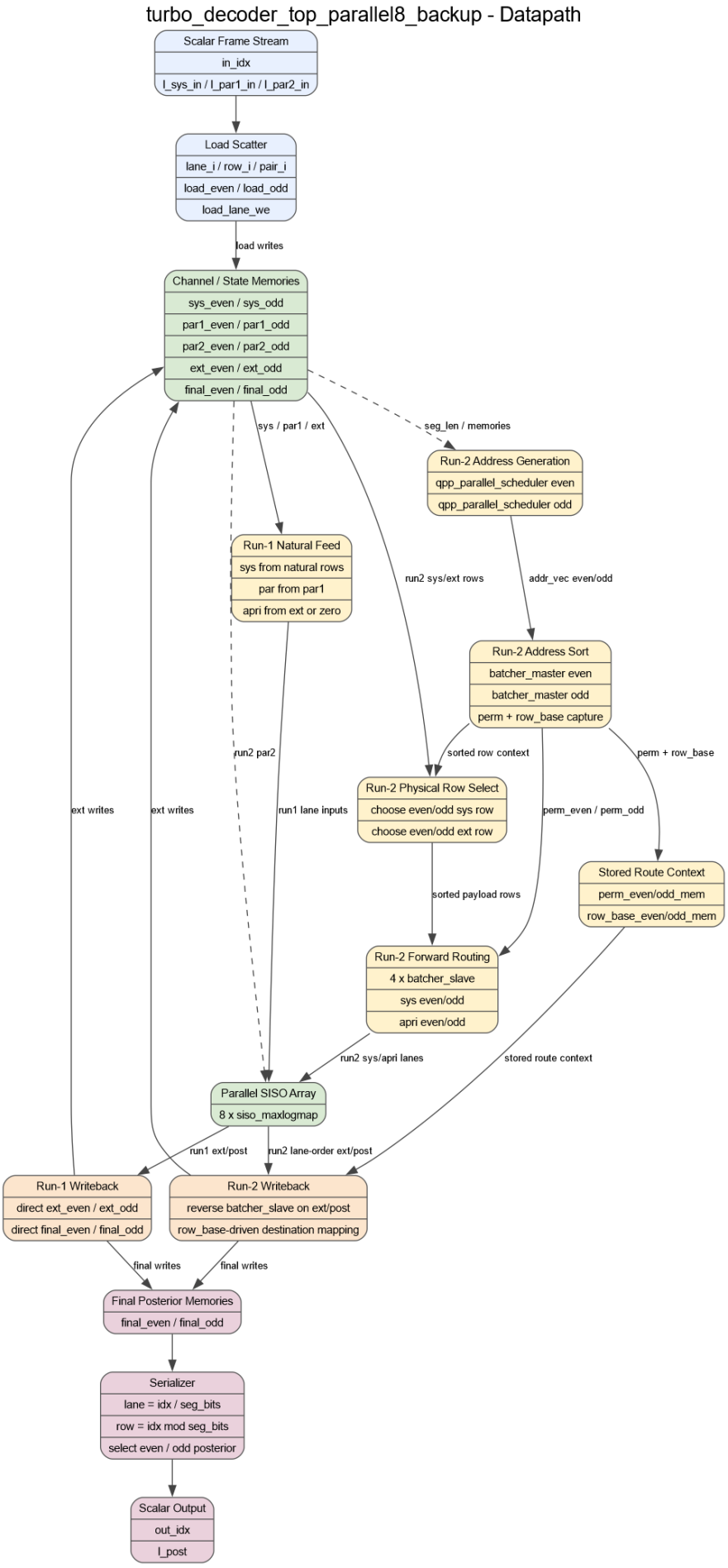
Batcher Slave

Slot Router Cell



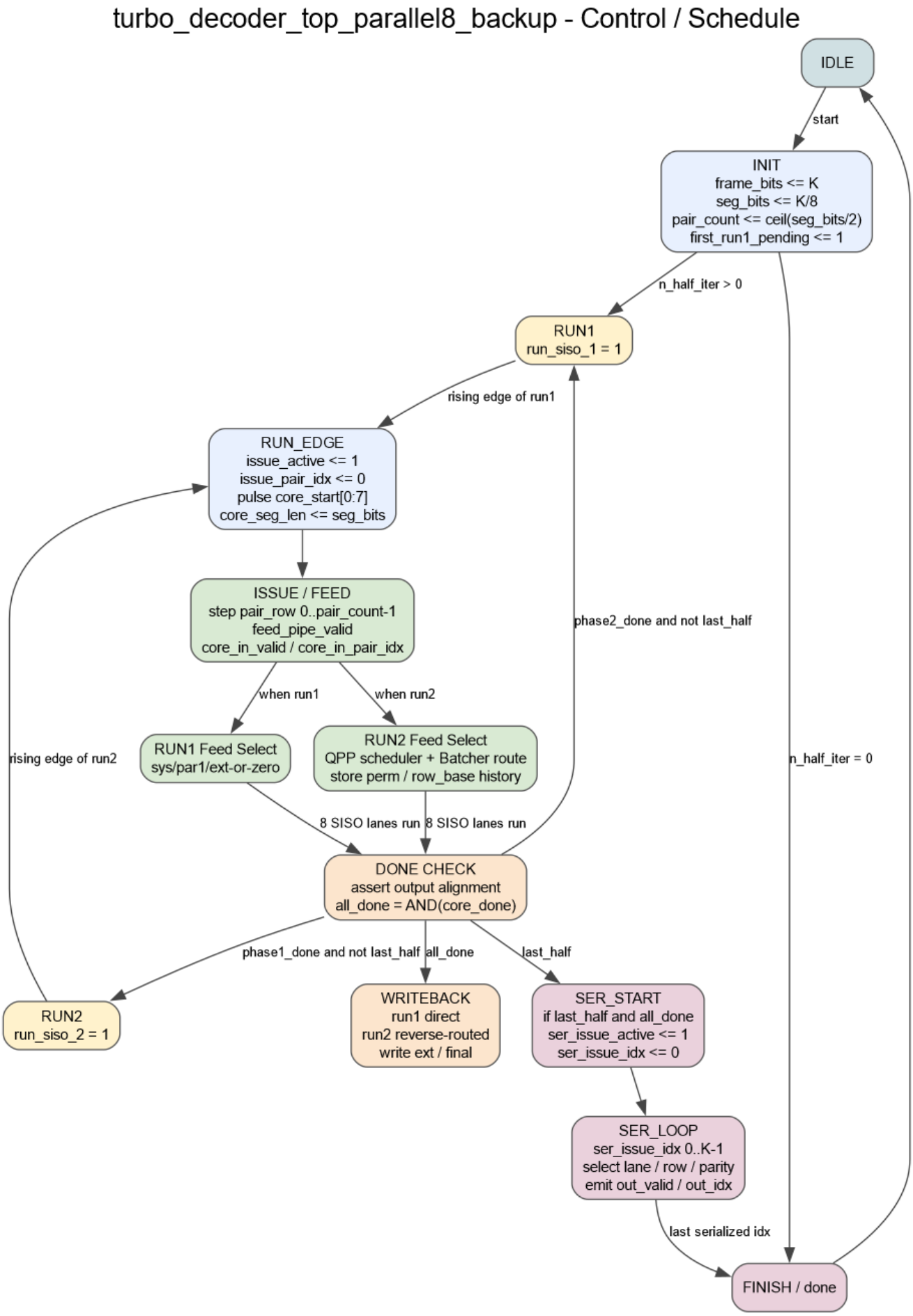
Parallel 8-SISO Top

Datapath



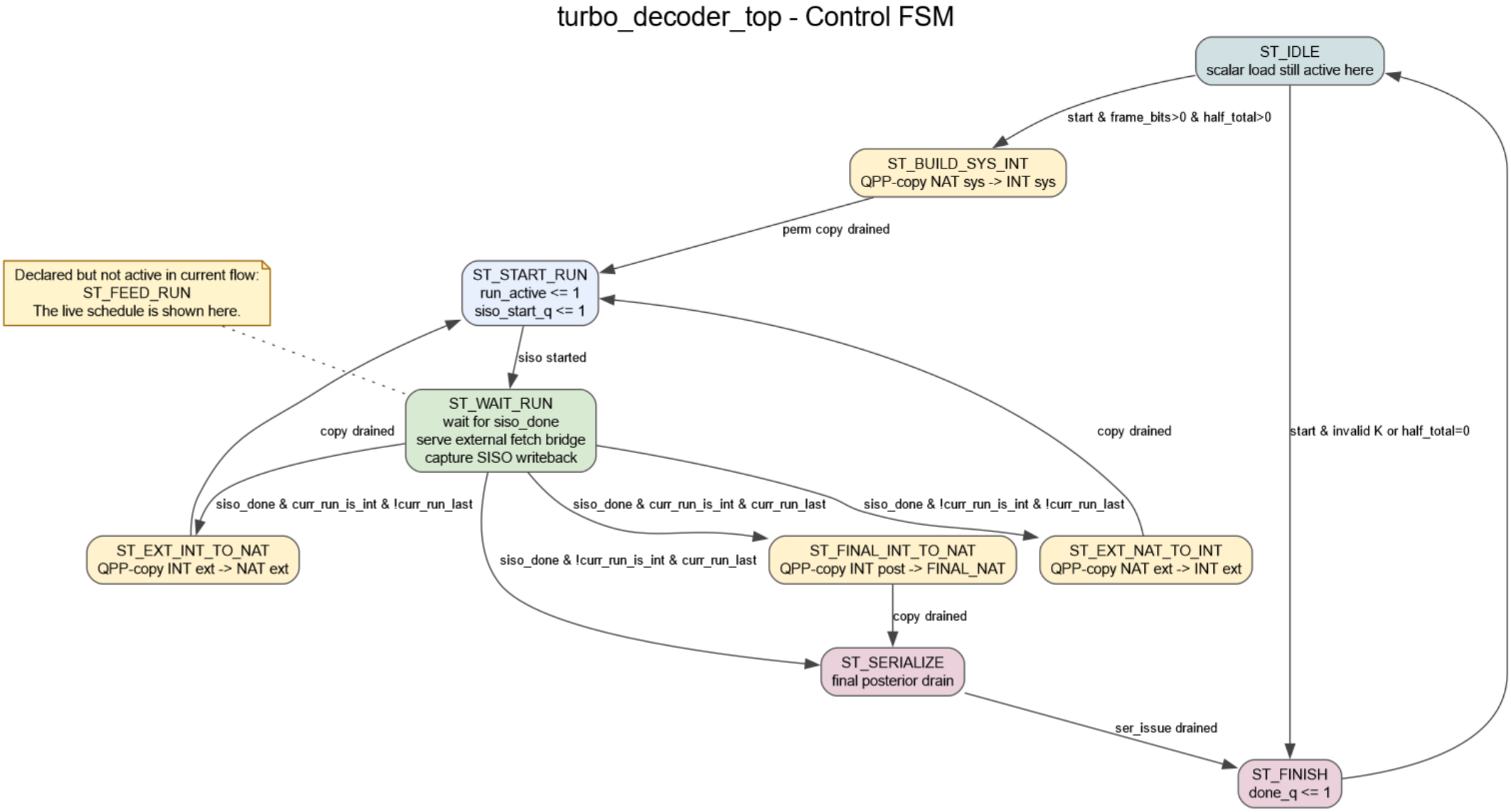
Parallel 8-SISO Top

Control Logic



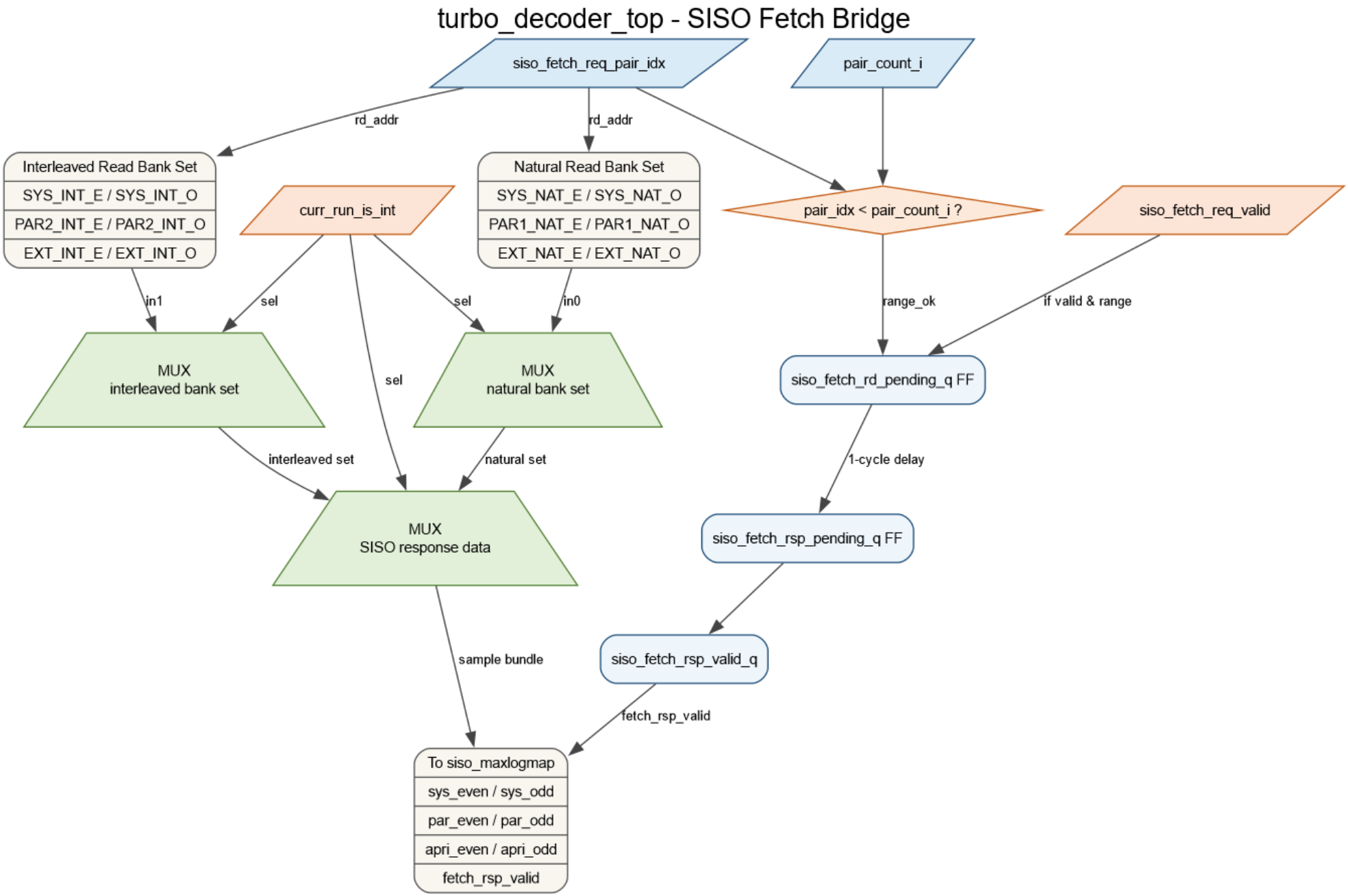
Active Single-SISO Top

Control FSM



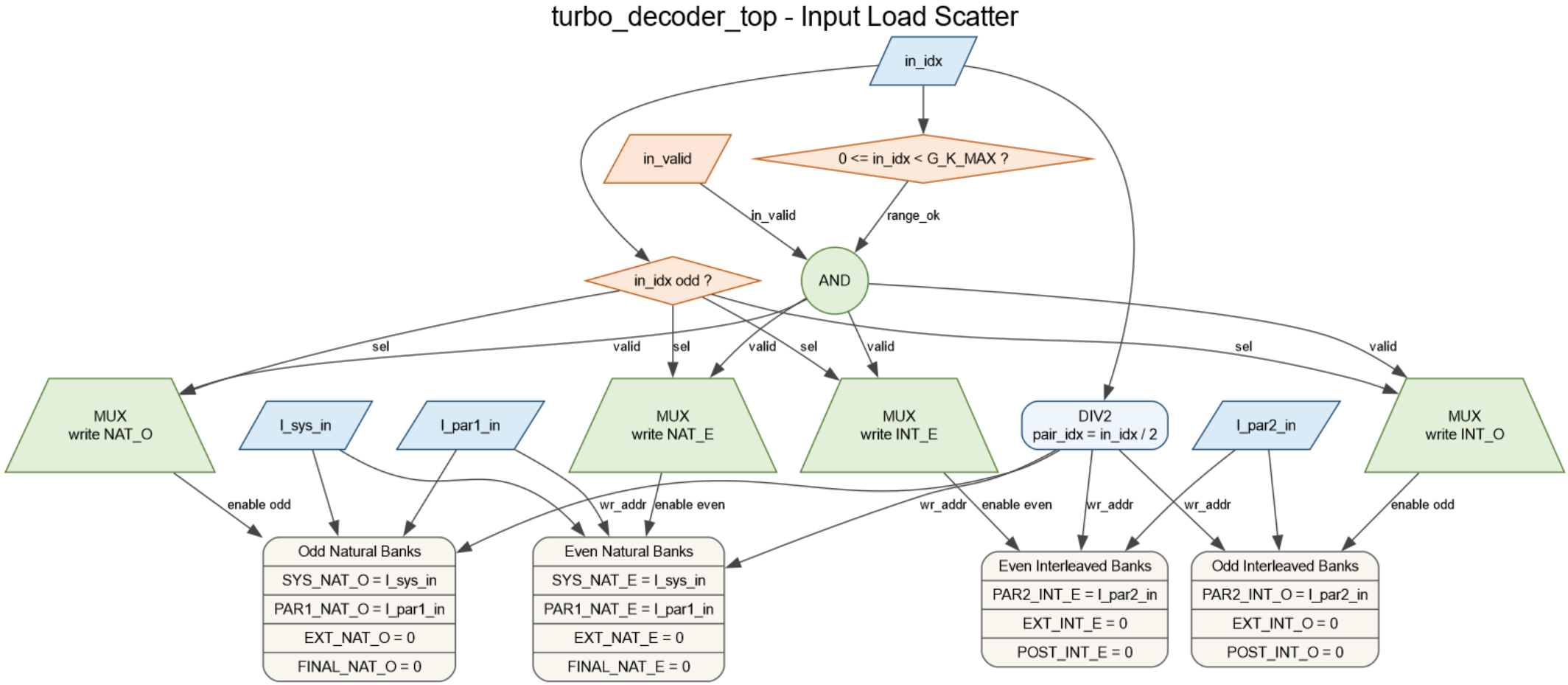
Active Single-SISO Top

SISO Fetch Bridge



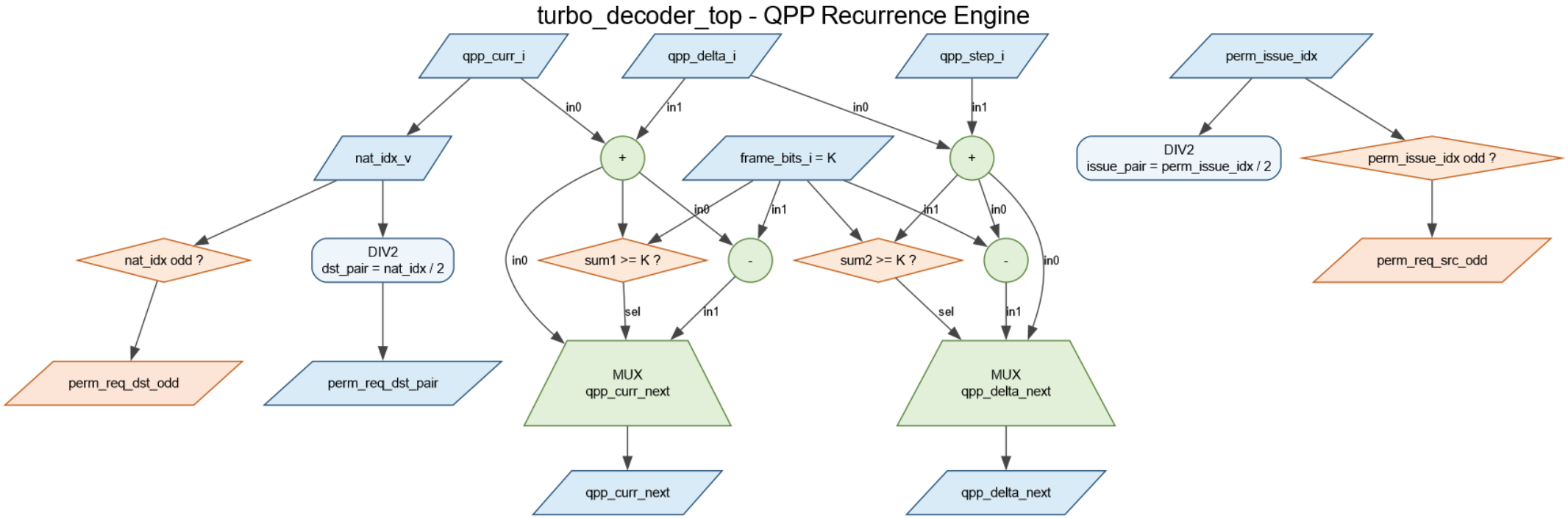
Active Single-SISO Top

Load Scatter



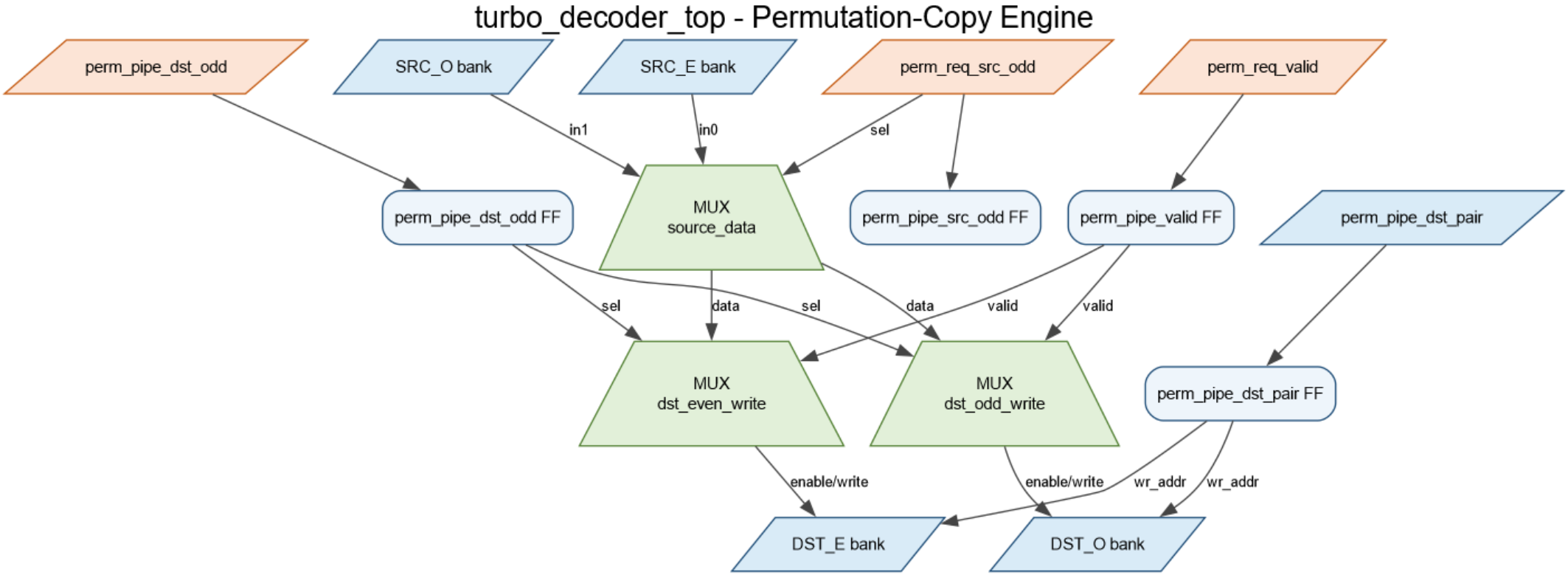
Active Single-SISO Top

QPP Recurrence



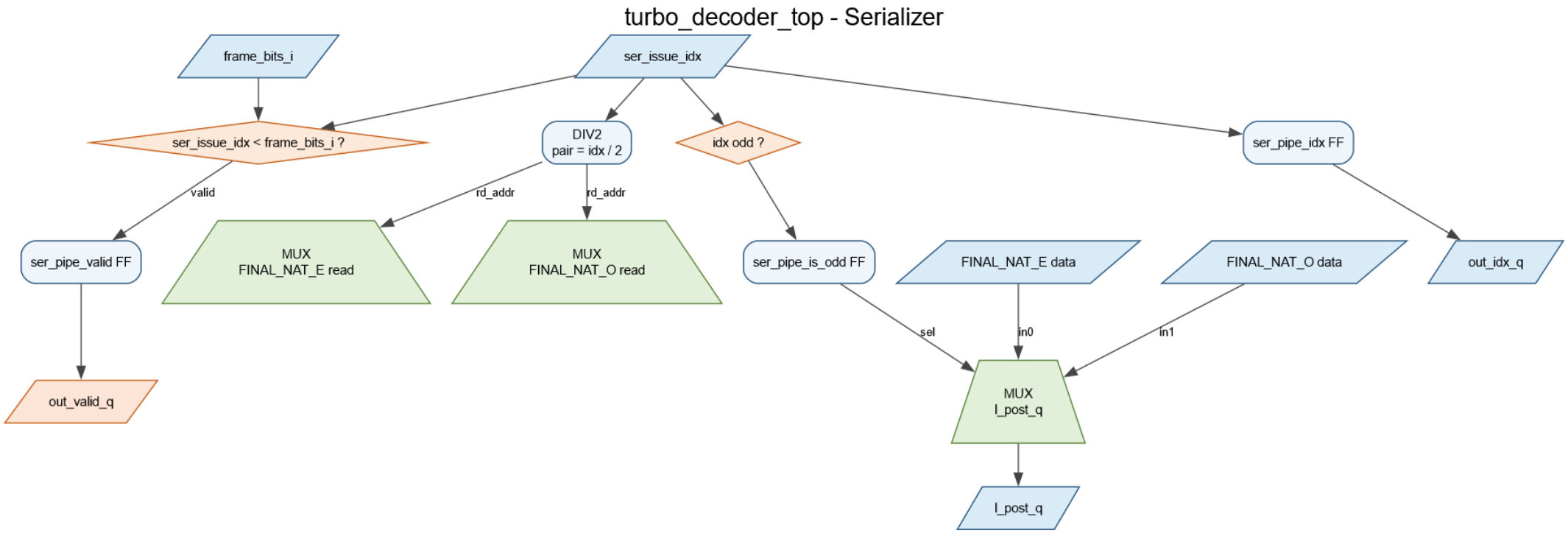
Active Single-SISO Top

Permute Copy



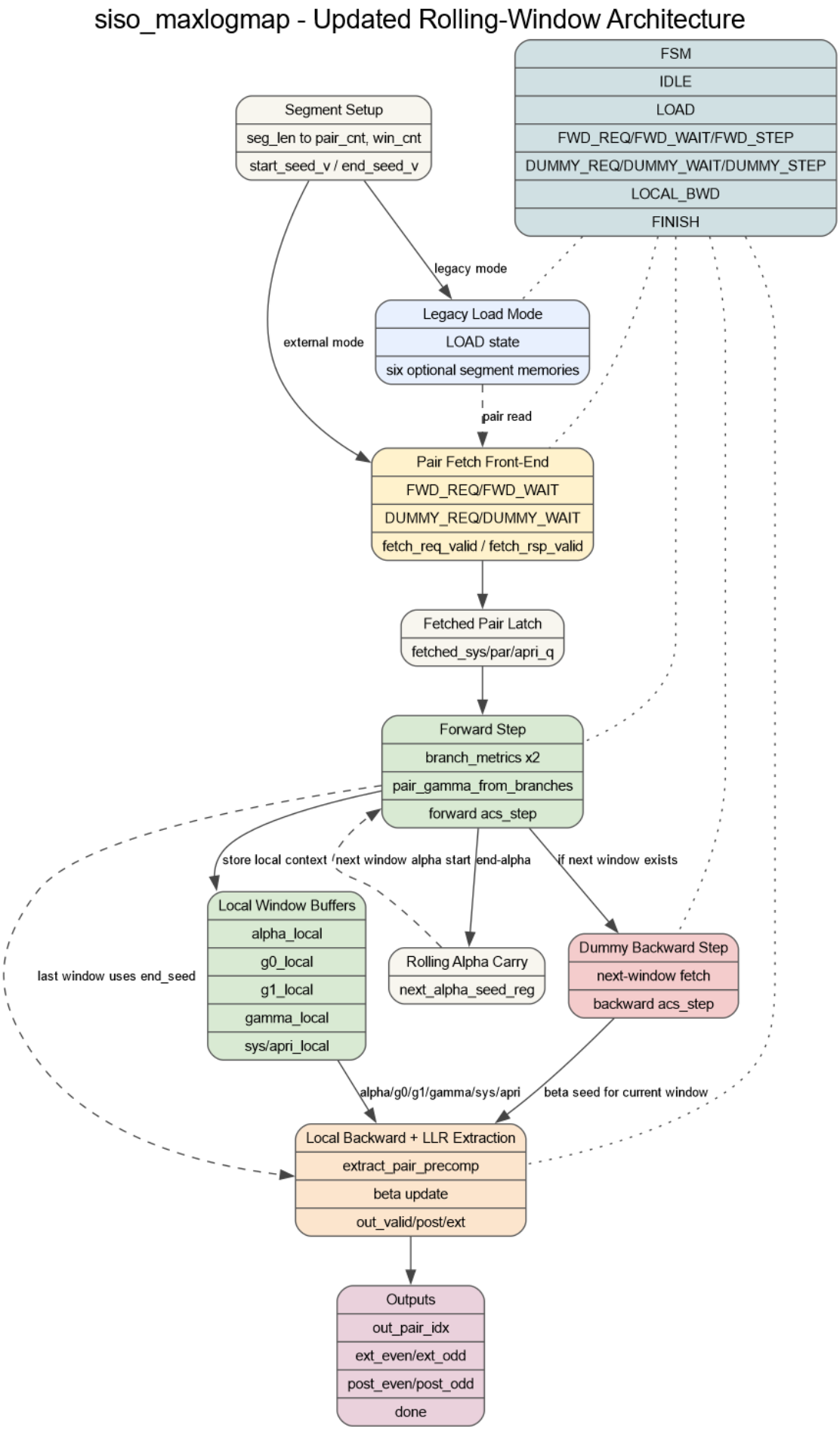
Active Single-SISO Top

Serializer



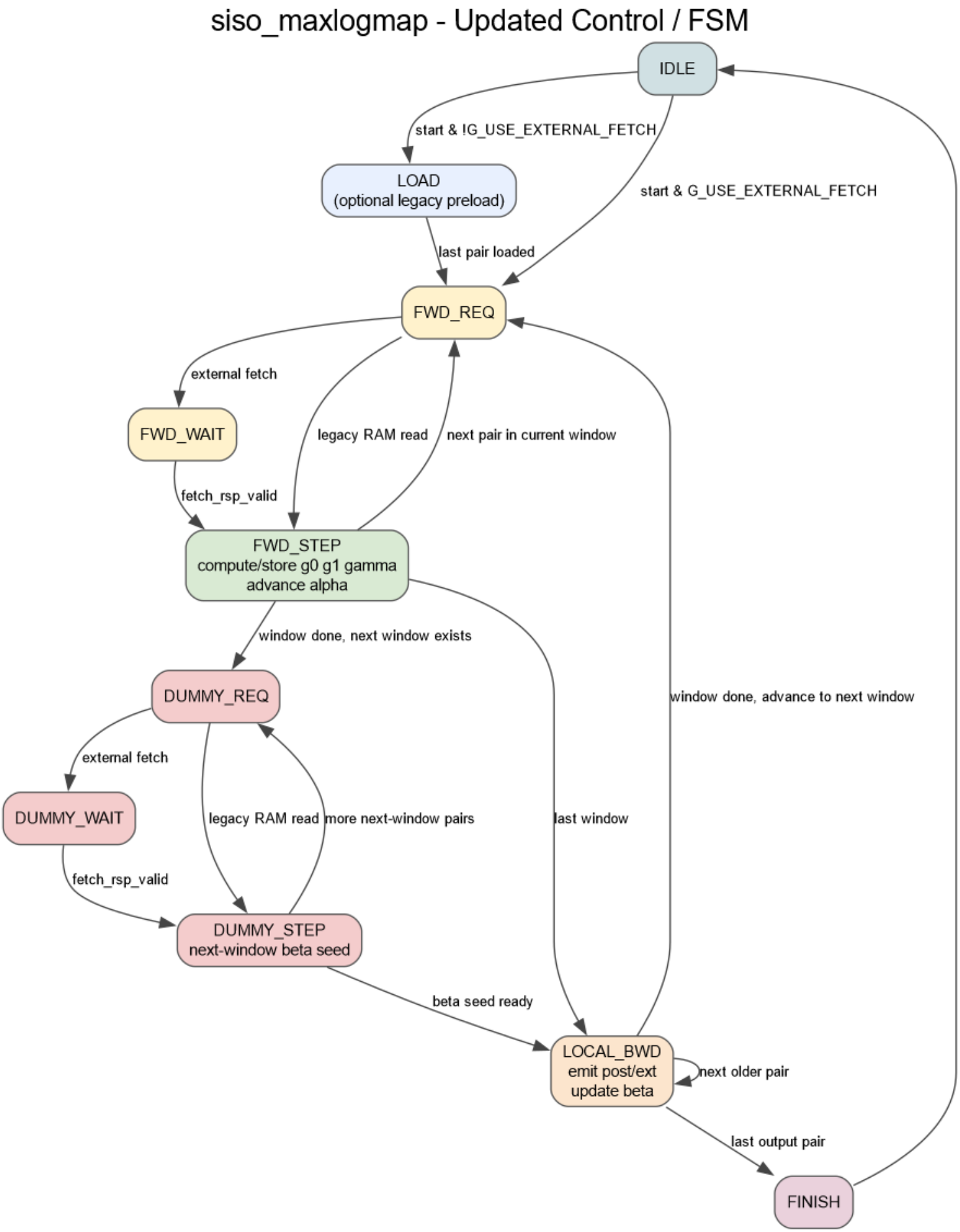
SISO Max-Log-MAP

Architecture



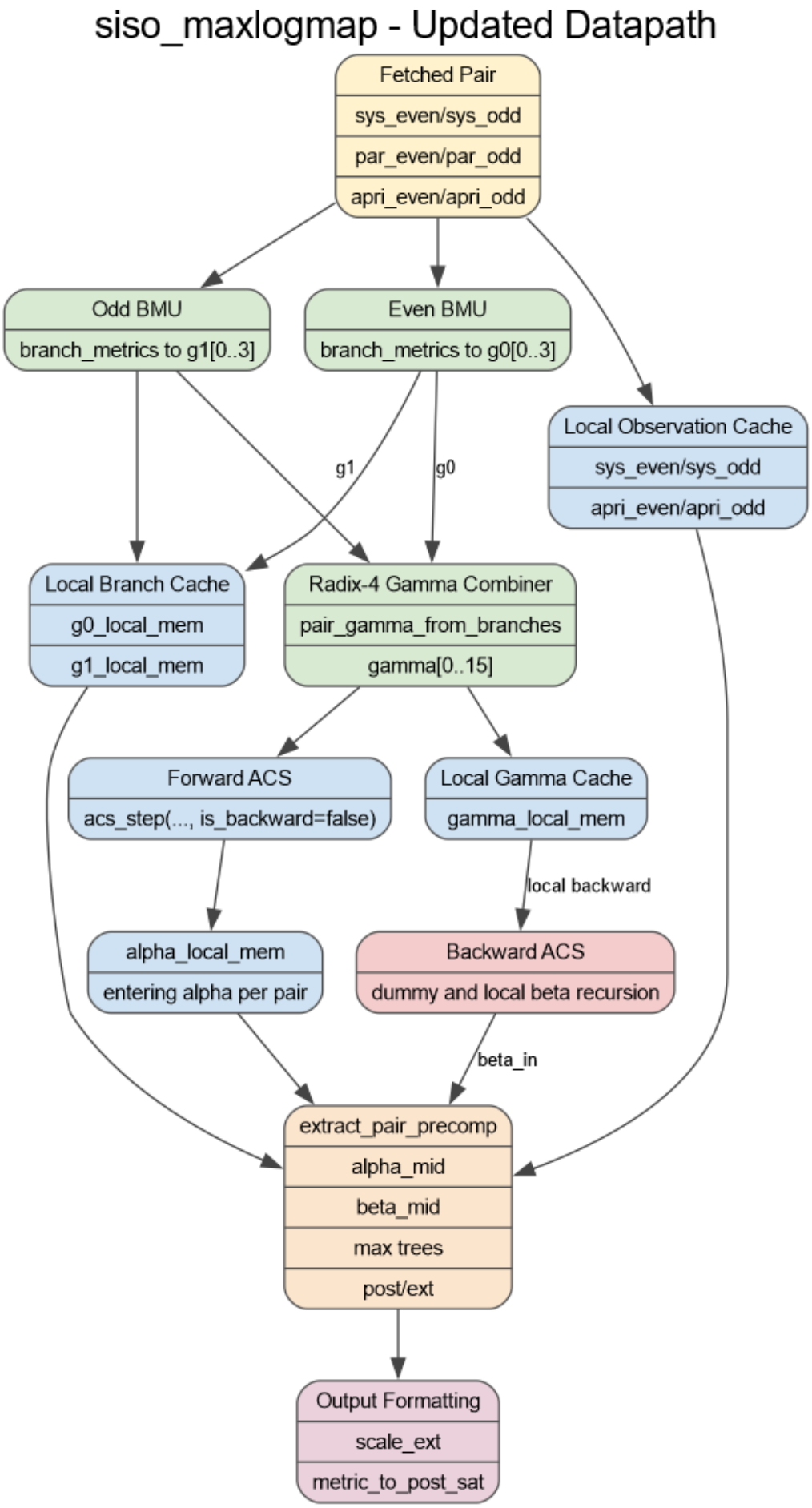
SISO Max-Log-MAP

Control



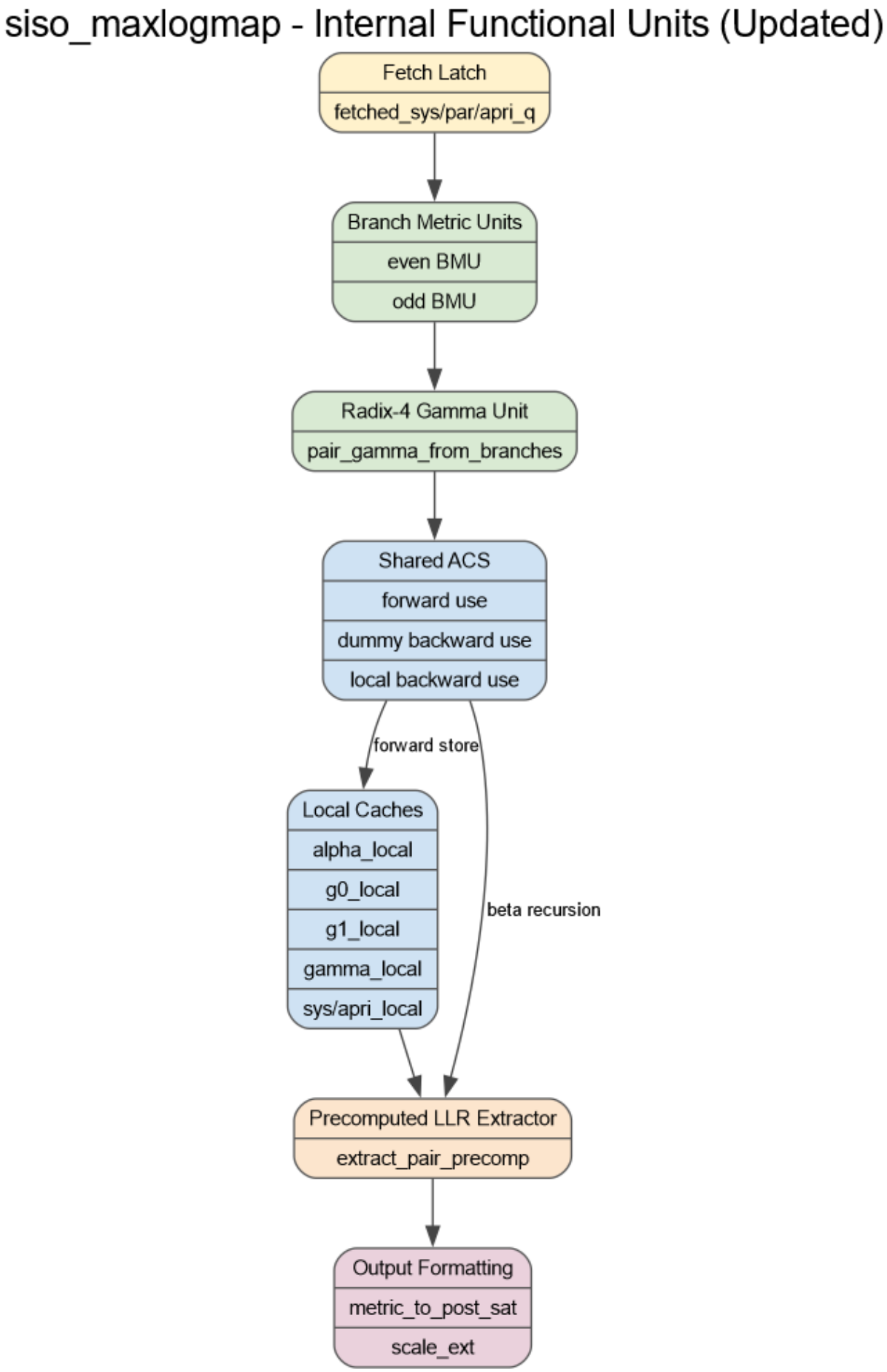
SISO Max-Log-MAP

Datapath



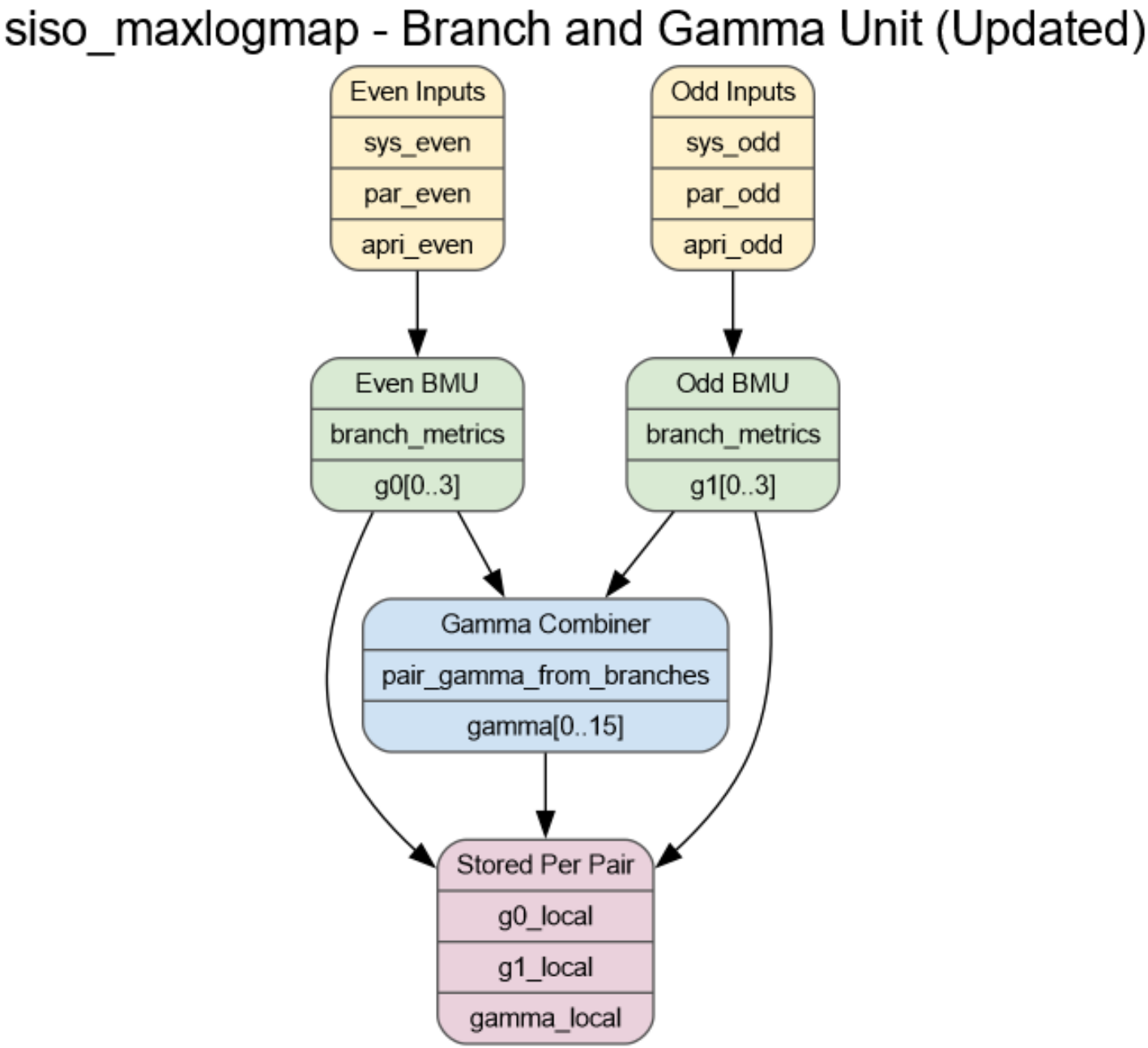
SISO Max-Log-MAP

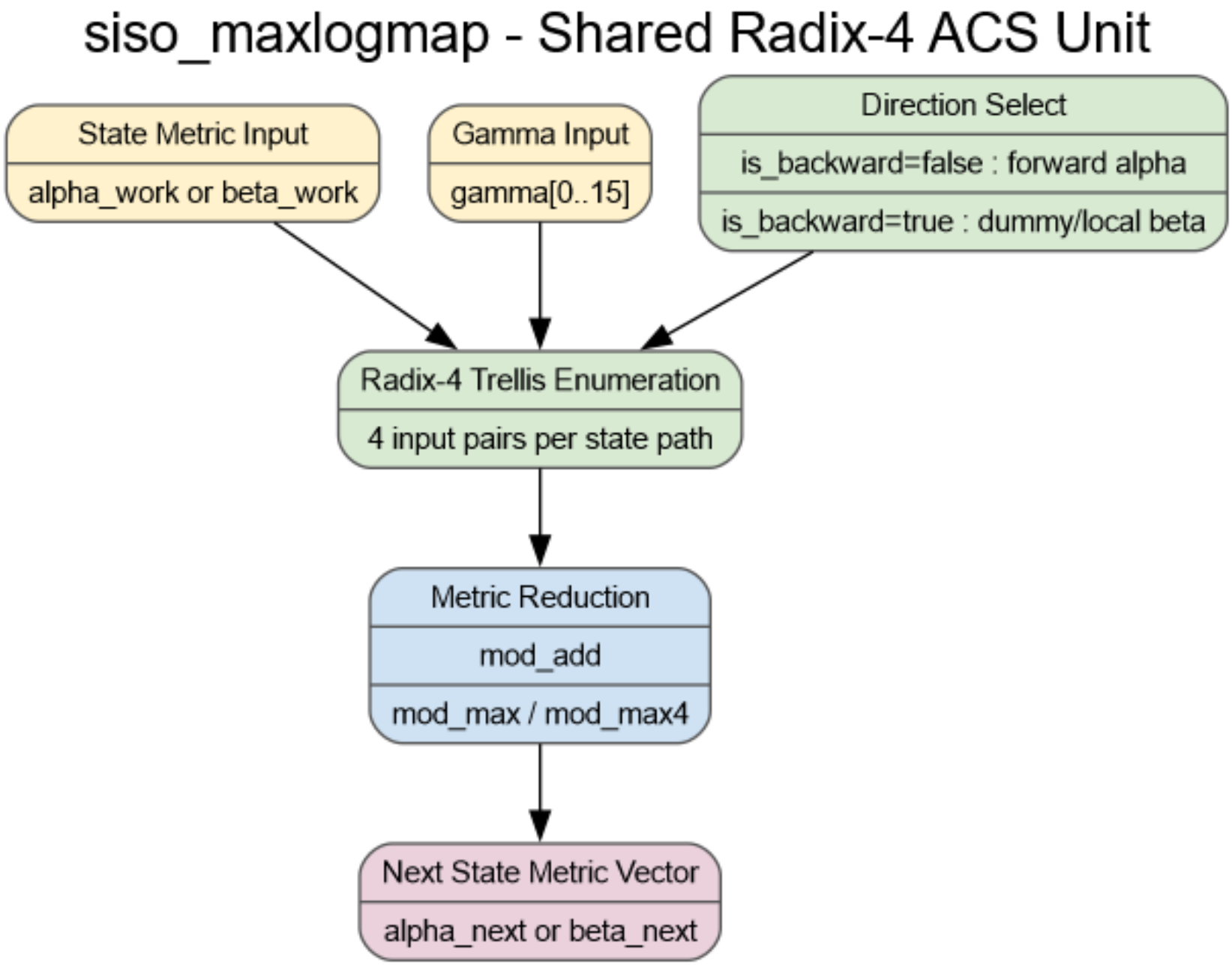
Functional Units



SISO Max-Log-MAP

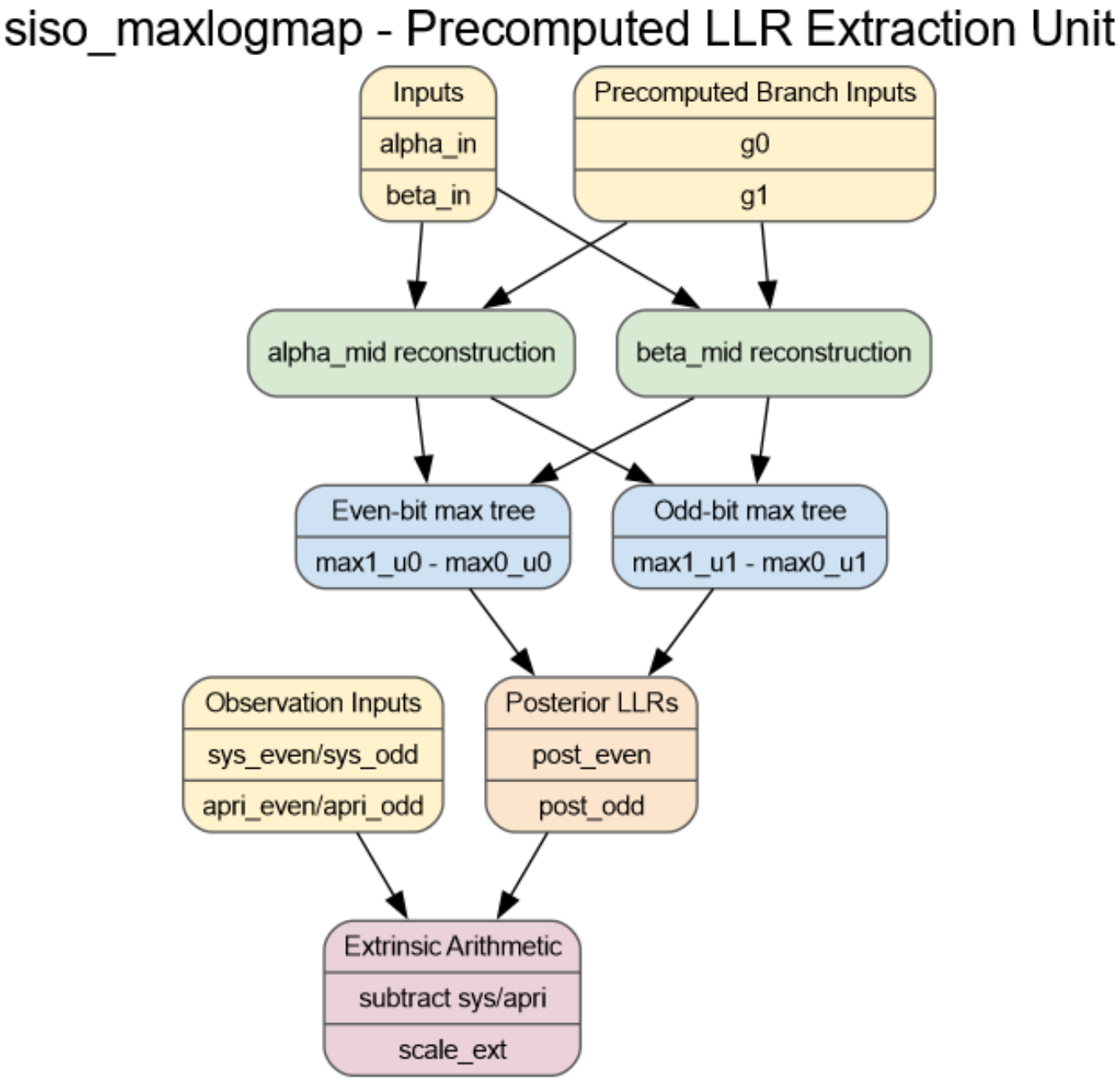
Branch and Gamma Unit




siso_maxlogmap - Shared Radix-4 ACS Unit
State Metric Input
alpha_work or beta_work
Gamma Input
gamma[0..15]
Direction Select
is_backward=false : forward alpha
is_backward=true : dummy/local beta
Radix-4 Trellis Enumeration
4 input pairs per state path
Metric Reduction
mod_add
mod_max / mod_max4
Next State Metric Vector
alpha_next or beta_next

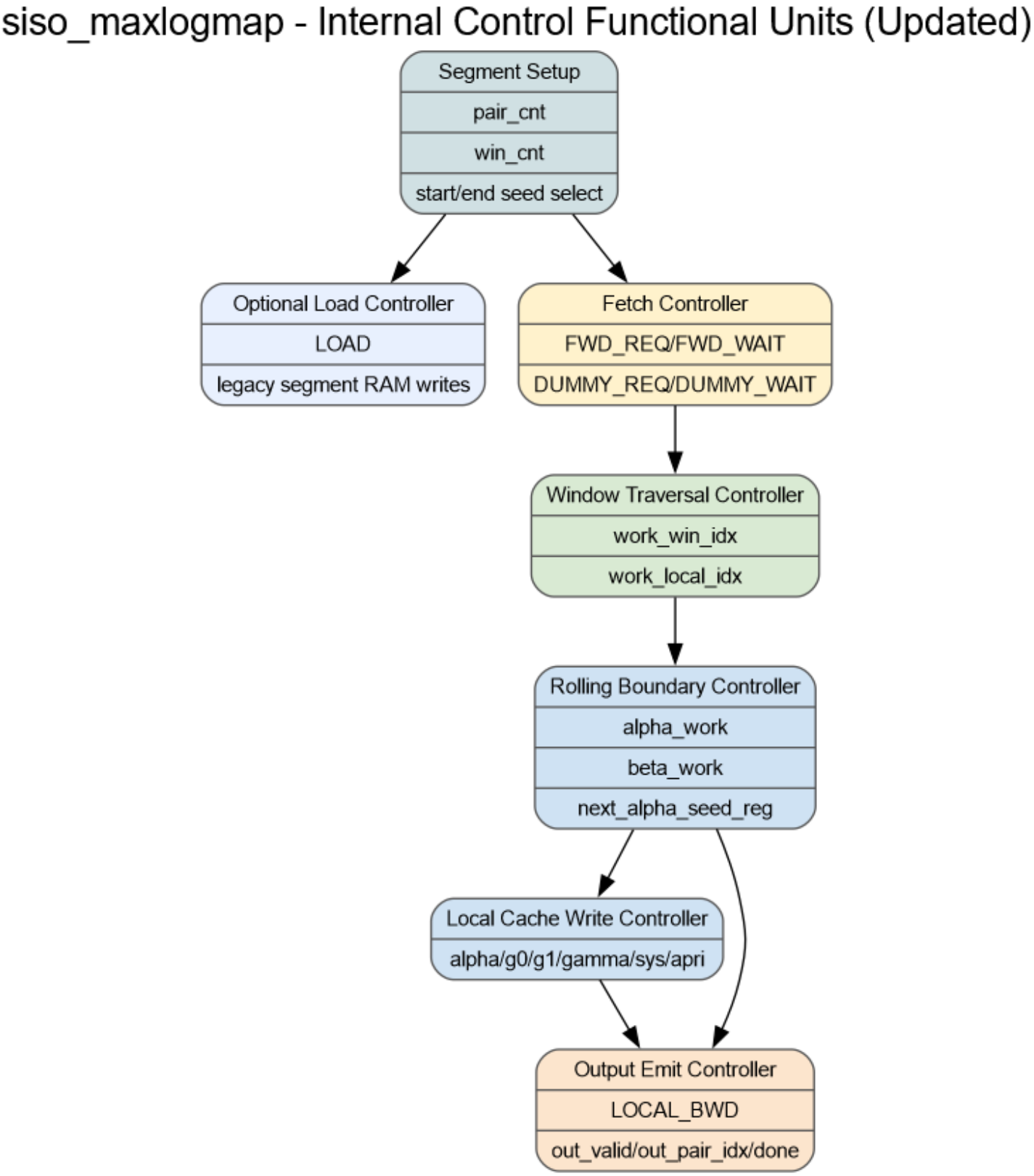
SISO Max-Log-MAP

LLR Extraction Unit



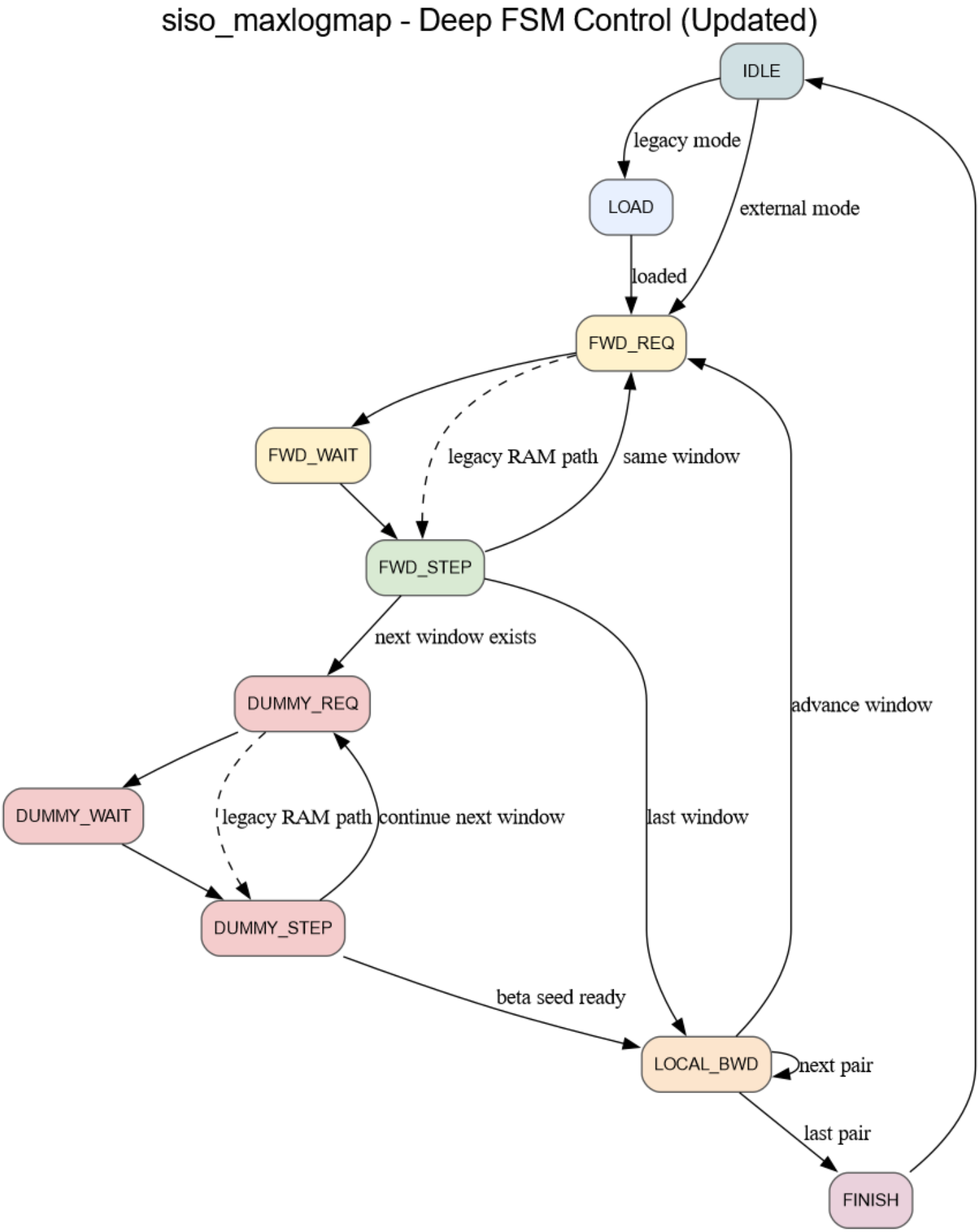
SISO Max-Log-MAP

Control Functional Units



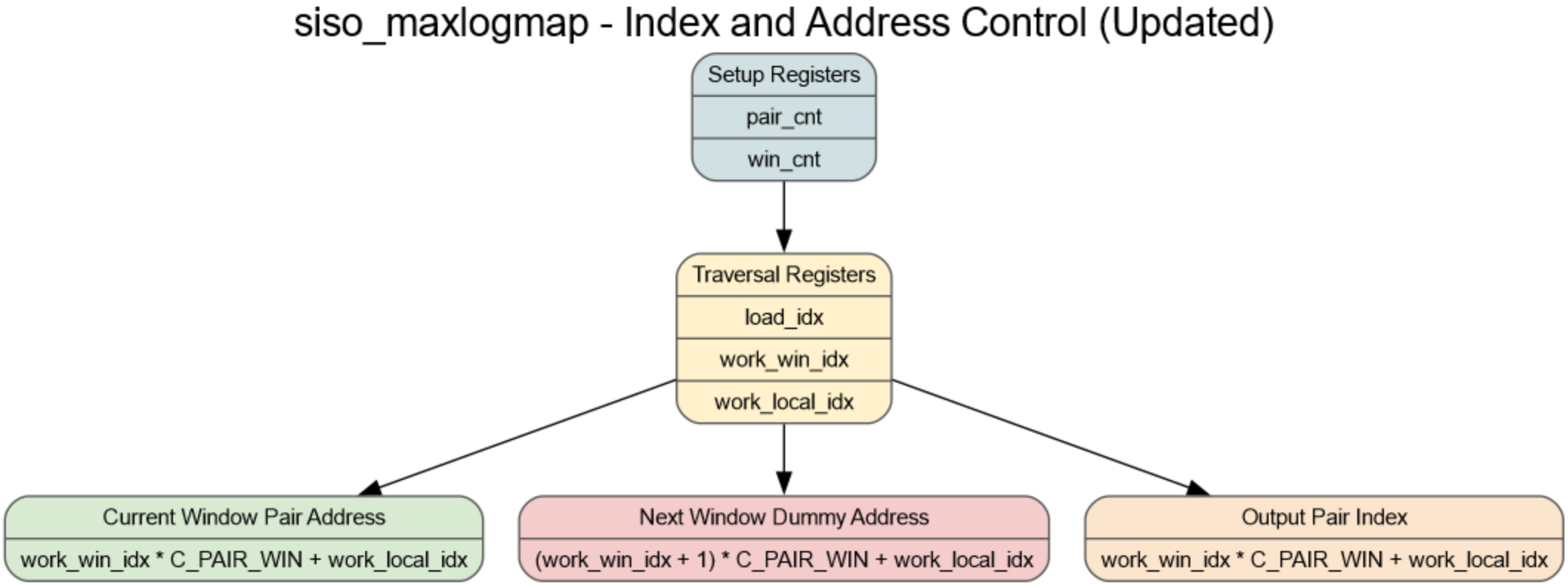
SISO Max-Log-MAP

FSM Deep Control



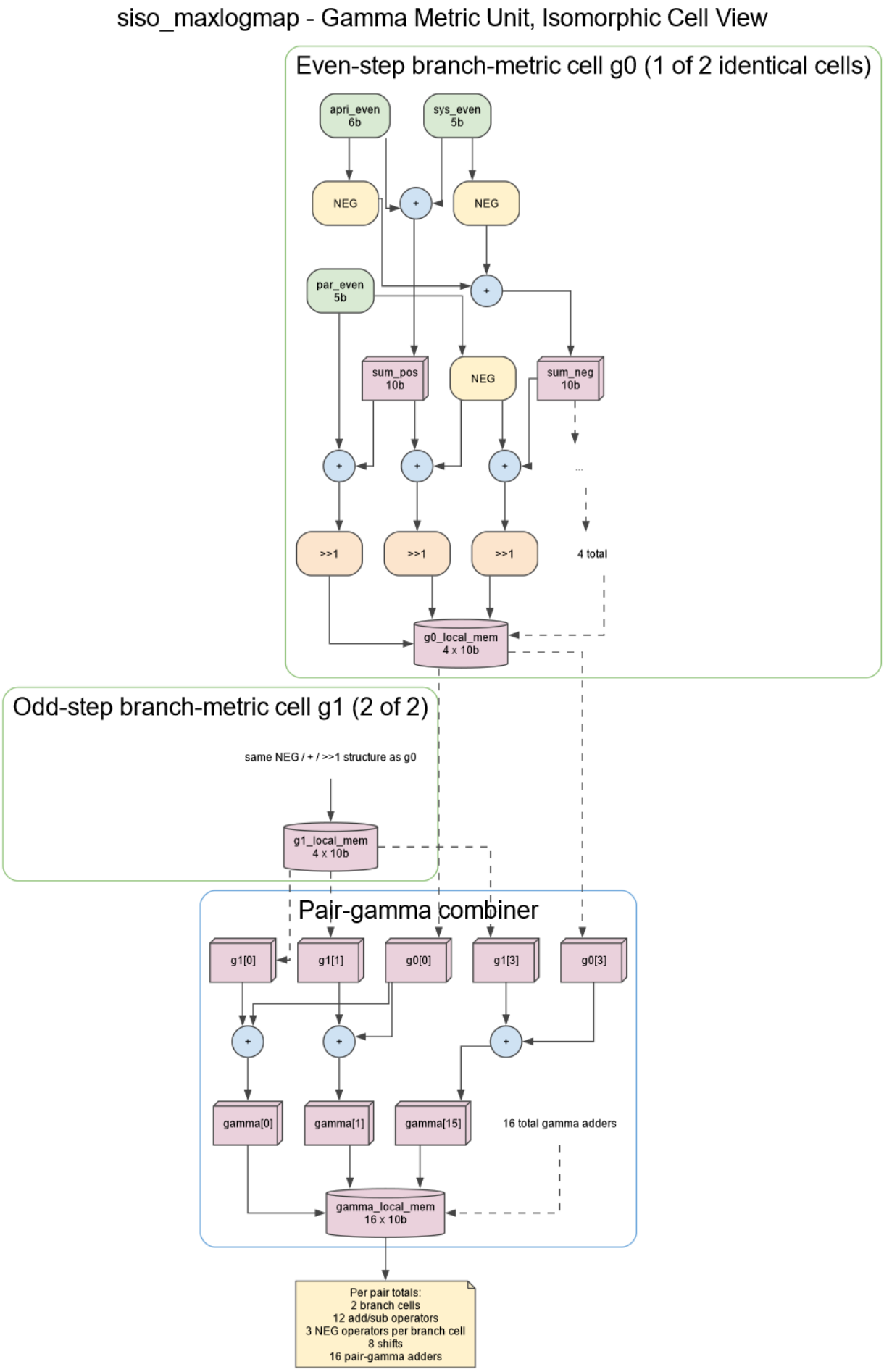
SISO Max-Log-MAP

Index and Address Control



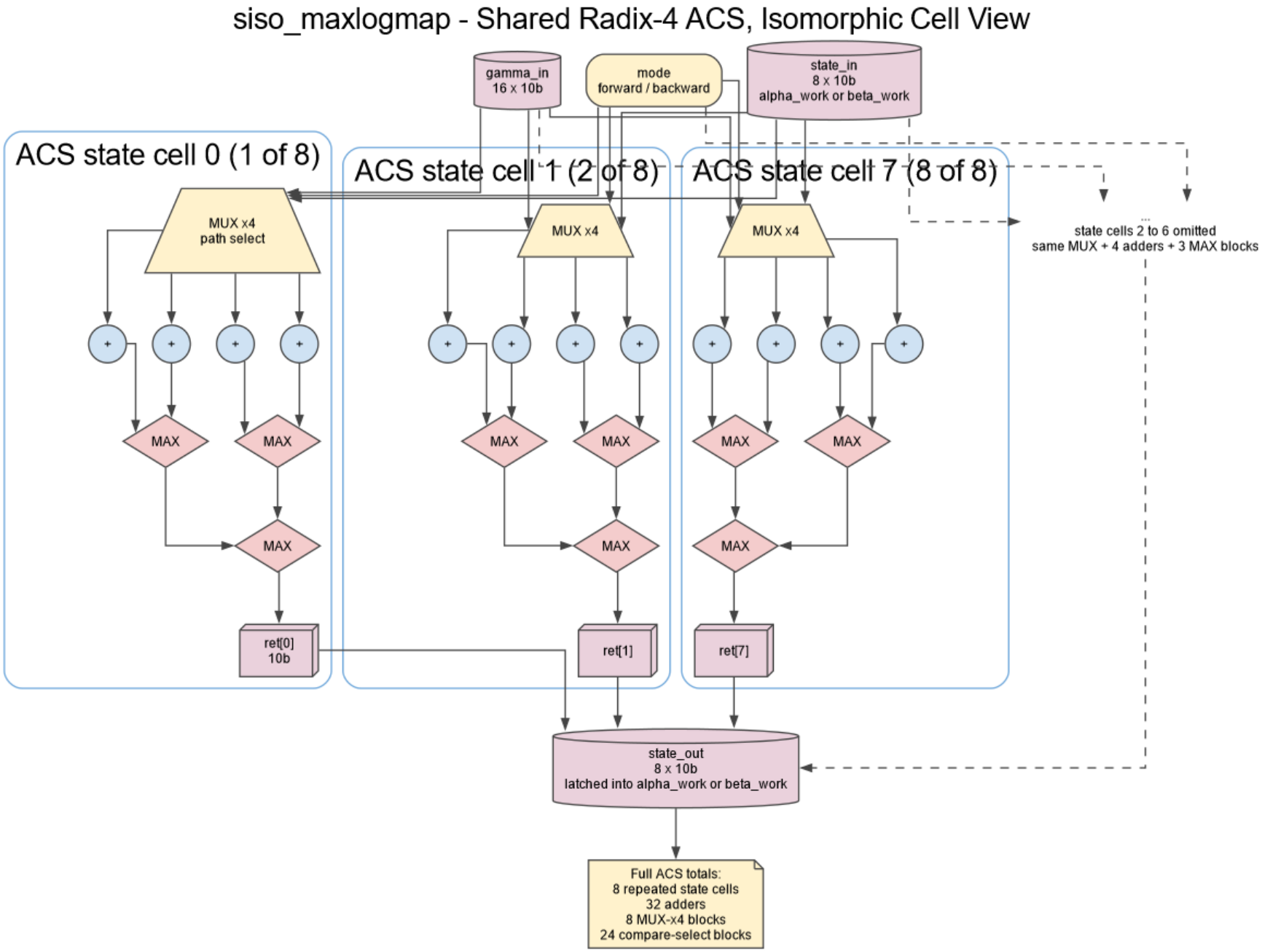
SISO Max-Log-MAP

Gamma Metric Cell

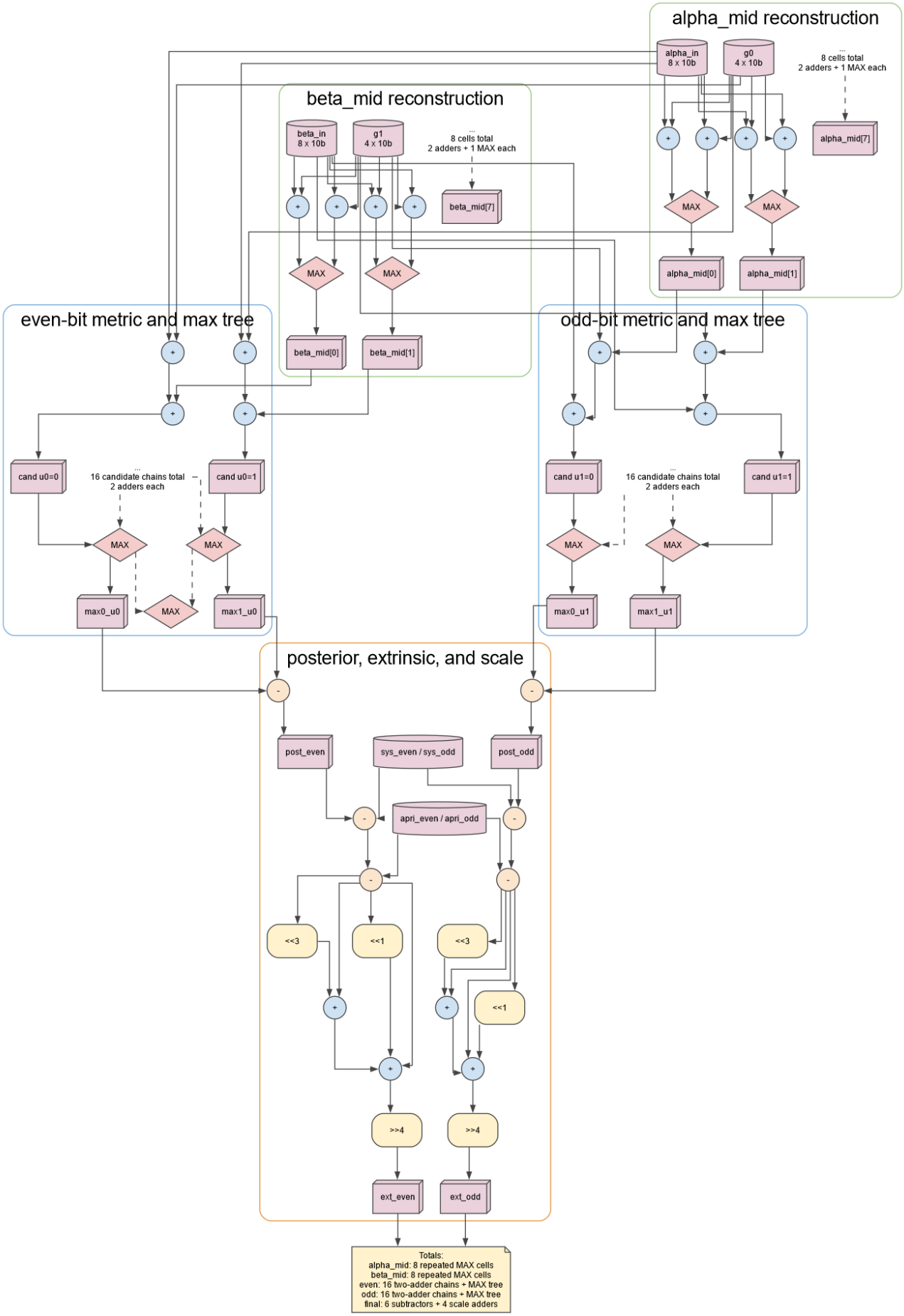


SISO Max-Log-MAP

Radix-4 ACS Cell



SISO Max-Log-MAP

LLR Extraction Cell

siso_maxlogmap - LLR Extraction, Isomorphic Cell View